

Northwind Aerial - Drone Operations Manual

1. Fleet Overview

Northwind Aerial operates a mixed fleet from its main hangar in Tacoma. The survey fleet consists of 42 NW-7 quadcopters, which handle roof inspections, crop mapping and short-range photography. Heavy-lift work is done by 6 NW-12 hexacopters, which carry sensor pods and small cargo loads up to the regulatory limit.

The NW-7 has a maximum takeoff weight of 2.4 kg including battery and camera gimbal. The NW-12 has a maximum takeoff weight of 11.8 kg and must only be flown by pilots holding the advanced endorsement. Each aircraft carries a serial plate under the battery bay, and that serial number is the key used in the maintenance log and in every flight record.

Aircraft are assigned to crews at the start of each week by the fleet coordinator. A crew may not swap aircraft with another crew without updating the assignment board, because firmware versions and calibration dates are tracked per airframe. Aircraft that are grounded for maintenance are tagged with a red streamer on the left front arm and moved to the quarantine shelf.

New pilots spend their first ten flights on the NW-7 under supervision before they are cleared for solo work. Supervision flights are logged with both the trainee and the supervising pilot named on the record.

2. Pre-flight Checks

Before every launch the pilot completes the pre-flight checklist in the crew app. The flight battery must read at least 95 percent charge before takeoff; a battery below that level is swapped, not topped up in the field. Propellers are inspected for chips, cracks and warping, and any damaged propeller is replaced as a full set on that motor pair.

Wind limits depend on the airframe. The NW-7 must not be launched when sustained wind exceeds 11 metres per second. The heavier NW-12 may be launched in sustained wind up to 15 metres per second, but gusts above that figure require the flight to be postponed. Pilots record the measured wind speed from the handheld anemometer, not the forecast.

The compass must be calibrated every 30 days, and additionally whenever the aircraft is moved more than 500 km from its last calibration site. GPS lock with at least 12 satellites is required before arming the motors. The return-to-home altitude is set 20 metres above the tallest obstacle in the operating area.

The visual observer confirms the launch zone is clear of people for a radius of 10 metres. If any checklist item fails, the flight is scrubbed and the failure is recorded with a photo in the crew app.

3. Battery Management

Lithium polymer batteries are the most common cause of drone incidents, so they are handled under strict rules. Batteries that will not be used within 48 hours are brought to storage charge, which is 3.85 volts per cell. The smart chargers in the battery room do this automatically when set to storage mode.

A battery is retired after 300 charge cycles, or immediately if it shows any swelling, puncture or a cell imbalance greater than 0.1 volts. Retired batteries are discharged fully in the salt-water bucket and sent to the certified recycler; they are never thrown in general waste.

Charging takes place only in the battery room, where the temperature must stay between 10 and 30 degrees Celsius. Every battery charges inside a fireproof bag, and a charger is never left running while the room is unattended. A class D extinguisher and a sand bucket sit by the door.

Each battery has a numbered label, and its cycle count is logged in the crew app at the end of every flying day. Batteries are rotated so that cycle counts across the pack stay roughly even.

4. Incident Reporting

Any incident, including a crash, a flyaway, a near miss with a manned aircraft, or injury to any person, must be reported to the safety officer within 2 hours of the event. The report is filed on form NW-IR-3 in the crew app, with photos of the scene and the aircraft.

In a flyaway, the pilot first attempts return-to-home, then switches to attitude mode and tries to regain manual control. If control cannot be regained, the pilot notes the last known position and heading and immediately notifies local air traffic control if the aircraft may enter controlled airspace.

Flight logs and telemetry for any aircraft involved in an incident are exported the same day and retained for 18 months. The aircraft itself is grounded until the safety officer signs off the investigation, and it may not be repaired before photographs of the damage are taken.

The safety officer reviews all incident reports at the monthly safety meeting, and lessons learned are added to this manual in the next revision.